

ALPINE, TX, USA

WMO: 720151

Lat: 30.383N

Lon: 103.683W

Elev: 4513

StdP: 12.45

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03049

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	23.8	-2.1	5.8	52.2	2.8	7.5	54.0	29.7	55.8	27.1	56.8	5.3	50	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	26.0	96.7	62.3	94.8	62.4	92.0	62.5	68.4	81.7	67.7	81.4	66.9	80.9	9.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	110.5	71.2	64.2	106.3	71.2	63.3	103.3	71.1	35.6	81.9	34.8	81.4	34.1	80.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.9	20.7	18.1	DB	13.8	100.3	4.9	2.1	10.3	101.8	7.4	103.0	4.6	104.2	1.1	105.7
			WB	12.9	70.5	4.5	0.6	9.6	70.9	7.0	71.3	4.4	71.7	1.2	72.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	45.4	50.5	57.1	63.9	71.1	78.3	77.2	76.3	70.7	63.6	53.3	47.1
	DBStd	13.83	9.20	10.27	8.91	7.42	7.31	5.03	4.40	4.70	5.53	7.82	8.97	9.46
	HDD50	599	199	108	39	4	1	0	0	0	0	10	71	167
	HDD65	2464	607	410	265	106	30	0	1	0	16	117	358	554
	CDD50	5321	57	122	259	420	655	850	844	815	622	430	169	78
	CDD65	1711	0	4	19	73	219	400	380	350	189	72	5	0
	CDH74	19499	7	115	446	1214	2865	4767	3904	3464	1725	862	119	11
	CDH80	8503	0	16	76	369	1324	2517	1811	1562	602	216	10	0
Wind	WSAvg	8.0	8.6	9.3	9.3	9.6	8.9	7.9	6.7	6.1	6.2	7.2	8.0	8.7
Precipitation	PrecAvg	16.1	0.5	0.5	0.4	0.5	1.1	2.3	3.0	3.0	2.8	1.5	0.6	0.6
	PrecMax	27.5	1.6	3.0	1.7	3.6	3.5	7.0	9.4	8.2	11.1	4.4	3.2	2.6
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0
	PrecStd	4.6	0.4	0.7	0.4	0.7	0.9	1.6	2.1	1.7	2.3	1.3	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	82.4	85.7	90.9	97.8	100.5	98.5	96.6	92.9	88.5	81.4	75.5
		MCWB	50.6	53.0	53.9	55.7	59.0	60.6	63.7	64.6	62.9	60.4	54.9	50.7
	2%	DB	70.7	77.6	82.1	87.4	94.2	97.9	94.8	94.1	90.0	85.5	77.4	72.2
		MCWB	48.5	50.9	52.3	53.9	58.3	61.7	64.1	64.2	62.1	59.6	54.2	49.8
	5%	DB	66.8	73.3	79.1	84.3	91.1	95.4	91.8	91.3	86.7	82.2	73.9	69.3
		MCWB	46.6	49.2	51.8	53.2	57.5	61.7	64.0	64.0	62.6	59.4	52.8	48.2
	10%	DB	62.4	69.9	76.1	81.4	88.3	92.6	90.2	89.4	83.4	79.2	70.4	63.9
		MCWB	44.5	47.9	50.2	52.7	56.9	62.7	64.3	64.7	62.1	58.5	51.0	46.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	55.7	59.5	61.9	67.7	69.5	69.3	69.3	68.8	67.4	59.6	53.9
		MCDB	69.0	75.0	72.4	75.6	80.2	83.0	82.7	81.4	79.7	80.1	70.2	69.5
	2%	WB	50.6	52.9	56.4	59.8	65.5	68.1	68.2	68.1	67.6	65.2	57.4	51.7
		MCDB	66.6	71.9	71.9	73.3	77.1	82.8	82.0	81.6	79.3	76.4	68.4	67.7
	5%	WB	48.0	50.8	54.3	57.8	64.0	67.1	67.3	67.2	66.7	63.6	55.3	49.7
		MCDB	64.6	70.0	71.3	73.1	76.4	82.4	81.5	81.4	78.0	74.5	69.2	65.4
	10%	WB	45.7	49.0	52.1	55.9	62.4	66.1	66.5	66.3	65.7	61.8	53.3	47.4
		MCDB	61.1	67.2	71.0	73.4	76.4	81.8	81.1	80.7	76.7	73.2	68.0	62.0
Mean Daily Temperature Range	5% DB	MDBR	25.6	27.9	28.6	29.2	27.6	26.0	23.1	22.7	22.3	25.7	26.9	25.7
		MCDBR	28.9	30.9	31.4	31.5	30.7	28.3	26.4	25.9	27.2	27.7	29.5	29.2
		MCWBR	14.3	13.8	13.3	13.0	11.2	8.5	7.0	6.6	8.1	10.1	13.2	14.3
	5% WB	MCDBR	27.4	28.8	27.5	28.2	27.6	25.6	22.4	21.7	20.3	22.5	26.1	27.5
		MCWBR	14.0	13.9	12.2	12.1	11.8	8.7	6.7	6.8	7.0	9.2	12.9	14.4
Clear-Sky Solar Irradiance	taub	0.249	0.256	0.269	0.300	0.317	0.347	0.369	0.349	0.331	0.288	0.265	0.251	
	taud	2.676	2.643	2.595	2.489	2.481	2.455	2.433	2.513	2.510	2.636	2.632	2.656	
	Ebn at Noon	320	325	325	315	307	296	289	295	299	309	311	314	
	Edh at Noon	23	27	30	35	35	36	36	33	32	26	24	23	
All-Sky Solar Radiation	RadAvg	1183	1491	1879	2224	2349	2330	2137	2007	1772	1541	1255	1094	
	RadStd	109	129	103	127	127	112	122	126	111	139	109	83	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+1.55	N/A	N/A	N/A	N/A	N/A	-313	+594	+383
(47) Regional (9 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185

Nomenclature: See separate page